The Mid-Cap Bridge: Enhancing Sub-Industry Resolution via Forced Connectivity and Universe Expansion

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Abstract

Financial embedding models often struggle to resolve fine-grained economic identities, frequently defaulting to broad sector classifications. Existing literature focuses on optimizing hyperparameters within fixed large-cap universes (S&P 500), typically achieving high hit rates only at the Sector or Industry group level. This paper demonstrates that expanding the universe to include mid-cap stocks (Russell 1000/3000), combined with a novel graph construction method and sufficient historical depth, significantly improves Sub-Industry Hit Rates (identifying peers within the exact same GICS Sub-Industry). We introduce a "Forced Connectivity" mechanism that applies strict correlation thresholds to remove noise, then repairs disconnected components by enforcing minimum degree $k = 3$, ensuring graph connectivity while preserving signal purity. Through systematic experimentation across 114 configurations spanning three universes, four training periods, and multiple Node2Vec parameters, we find that universe expansion combined with 5-year training periods yields hit rates surpassing 56% for top-3 neighbors at the 163-class GICS Sub-Industry granularity—substantially above the approximately 2% random baseline and representing meaningful progress on an underexplored fine-grained classification task. Our best-performing configuration employs an intermediate exploration parameter ($q = 7.2$), suggesting that balanced random walk strategies outperform both extreme depth-first and breadth-first approaches. We find that mid-cap stocks correlate with improved embedding resolution, potentially acting as topological intermediaries, though this effect is partially confounded with the benefits of extended training periods and improved covariance estimation from larger sample sizes.

1 Introduction

The classification of financial assets has traditionally relied on manual taxonomies such as the Global Industry Classification Standard (GICS), which often lag behind the dynamic shifts of modern markets [23]. Recent advances in Graph Representation Learning have attempted to automate this process through embedding techniques [3, 1, 9].

We constructed graphs where nodes represent individual stocks, connected by edges weighted according to correlation strength between the movement of every stock pair. Using the Node2Vec algorithm—inspired by Word2Vec from natural language processing—we generate multidimensional vectors through weighted random walks [2]. Essentially, a virtual "walker" randomly visits 20 nodes that are directly connected to one another 100 times for every node. The walker traverses these edges according to parameters p (return parameter) and q (in-out parameter). High p discourages immediate backtracking, while q controls local versus global exploration: low q encourages distant exploration (Depth–First Search or DFS), high q promotes local neighborhood saturation (Breadth–First Search or BFS).

The seminal work by Sarmah et al. (2022) at BlackRock applied Node2Vec to S&P 500 correlation networks to replicate GICS sectors [1]. Their model successfully demonstrated that non-linear relationships could be compressed into embeddings using Minimum Spanning Tree (MST) filtering. However, we identify a fundamental limitation: by forcing every node into a single connected component with minimal edges, MST breaks valuable connections only because it is not the strongest connection for that given node. In actual markets, stocks are deeply interconnected [14, 15], suggesting that a node in our graph should have a greater degree than one or two like it does in an MST.

Our research improves upon existing models through:

1. **Forced Connectivity Repair:** We apply strict correlation thresholds that allow graph fragmentation, then repair connectivity by enforcing minimum degree $k = 3$, reconnecting isolated nodes to their strongest correlation partners without compromising signal quality.

2. **Reliance on Simple Correlation:** Although we test filtering techniques such as Graphical Lasso [10], our most successful networks rely on simple Pearson correlation between stock returns. This yields both a healthier graph and superior hit-rates, consistent with findings that sparse precision matrix estimators can produce unstable networks in financial applications [16].

3. **Granularity Shift:** We utilize GICS Sub-Industries (163 classes) as our primary ground truth, rather than broad sectors (11 classes), representing a 14× finer classification challenge.

4. **Extended Training Periods:** We discover that 5-year training windows significantly outperform shorter periods, consistent with portfolio optimization literature suggesting that stable covariance estimation requires multi-year samples [12, 19].

5. **Systematic Parameter Exploration:** We explore Node2Vec parameters across a wide range ($q \in [0.5, 20.0]$), finding that intermediate values ($q = 7.2$) achieve optimal performance by balancing local and global exploration.

2 Related Work

2.1 Correlation Networks in Finance

The study of correlation-based financial networks has a rich history dating to Mantegna's seminal work on hierarchical structures [6]. Researchers have employed various filtering approaches including MST, Planar Maximally Filtered Graphs (PMFG) [7], and threshold methods [13, 14]. Esmalifalak (2025) provides a comprehensive review of correlation network methodologies in finance [17]. A persistent challenge in this literature is the threshold-connectivity tradeoff: strict thresholds remove noise but fragment the network, while lenient thresholds preserve connectivity but include spurious correlations [13].

2.2 Stock Embeddings and Industry Classification

Early work by Gopikrishnan et al. (2001) demonstrated that business sectors could be identified from stock price fluctuations alone [21]. More recently, Dolphin et al. (2022) proposed neural models for learning stock embeddings from returns data, achieving approximately 60% accuracy at the 11-sector level [3]. Subsequent work has explored multimodal approaches combining price data with news [4] and contrastive learning frameworks [5]. Wang et al. (2020) proposed Stock2Vec using temporal convolutional networks [22]. Our work differs by targeting the significantly harder 163-class sub-industry classification task and introducing a novel graph construction method.

3 Methodology

3.1 Data

We utilized daily adjusted closing prices for constituents of the S&P 500, Russell 1000, and Russell 3000. All price data was acquired from Massive.com (formerly Polygon.io). Data was filtered to include only stocks with full trading history (100% coverage) within each training window. The primary evaluation metric is the Sub-Industry Hit Rate, which measures the accuracy of the top-3 nearest neighbors against the GICS Sub-Industry classification obtained from Russell 3000 ETF holdings data.

Training periods tested: 6 months, 1 year, 2 years, and 5 years, with all windows ending at the data cutoff date. Returns were calculated as daily percentage changes, with standard preprocessing to handle missing data through forward-filling (e.g. filling in data for assets with ticker changes or mergers) and removal of flat-line stocks.

3.2 Graph Construction and Forced Connectivity

A critical innovation in our workflow is the handling of graph topology through a novel two-stage process we term "Forced Connectivity." This approach addresses the well-known threshold-connectivity tradeoff in correlation network construction [13, 17].

Stage 1: Strict Thresholding. We first compute the Pearson correlation matrix from daily returns and apply a strict threshold (typically $t > 0.65$). Only stock pairs with absolute correlation exceeding this threshold receive edges in the initial graph:

$$G_0 = \{(i, j) : |\rho_{ij}| > t\} \tag{1}$$

This aggressive filtering effectively removes weak correlations and market-wide noise, but typically results in isolated nodes or small disconnected components ("islands"), a known consequence of threshold methods [14].

Stage 2: Minimum Degree Enforcement. Rather than lowering the threshold or forcing a tree structure, we implement a targeted repair mechanism. For every node v in the graph, we enforce a minimum degree constraint:

$$\deg(v) \geq k = 3 \tag{2}$$

If a node has fewer than 3 connections after thresholding, we forcefully reconnect it to its k strongest correlation partners (by absolute correlation magnitude), regardless of whether those correlations exceed the threshold. This ensures that:

- Every isolated island reconnects to the global manifold which allows for Node2Vec traversal
- Nodes retain connections to their most correlated peers through high-signal edges from Stage 1
- Only the most important weak connections remain in the graph
- Nodes are connected to many other nodes just like real stocks

This approach differs fundamentally from MST methods [6, 7]: rather than imposing a global tree structure that connects all nodes with exactly $n - 1$ edges, we selectively repair only problematic nodes, preserving the dense local clusters created by strict thresholding while ensuring navigability. Similar hybrid approaches combining filtering methods have shown promise in recent work [18].

Computational Details: The repair mechanism operates iteratively. For each under-connected node, we identify its strongest remaining correlation partner (excluding nodes already connected and itself), add that edge, and repeat until $\deg(v) \geq 3$. The correlation matrix is reused from Stage 1, avoiding redundant computation. In practice, this adds 30-80 edges to graphs of 500-2500 nodes, representing a 1-3% increase in edge count.

3.3 Alternative Filtering Methods Tested

We compared our threshold-plus-repair approach against two alternatives:

Graphical Lasso computes a sparse precision matrix (inverse covariance) using ℓ_1 regularization [10], theoretically removing indirect correlations and isolating direct relationships. However, sparse precision matrix estimators have been noted to produce unstable networks in financial applications [16]. In our experiments, Graphical Lasso-based graphs consistently underperformed simple correlation thresholding by approximately 5 percentage points in hit rate.

MST guarantees connectivity by constructing a tree with exactly $n - 1$ edges connecting all nodes [6]. While elegant, MST breaks meaningful connections between strongly correlated nodes solely to maintain tree structure as an MST, by definition, cannot contain more than N-1 edges and prevents the formation of cycles. These are neither a requirement for Node2Vec to work, nor are they representative of relationships between assets in actual stock markets [8, 15].

3.4 Embedding Mechanics: Node2Vec Parameters

For the Node2Vec parameters, we systematically tested $q \in \{0.5, 7.2, 10.0, 20.0\}$ with $p \in \{0.8, 1.0, 5.0\}$. Each configuration used 100 walks of length 20 per node, with embedding dimensions varying from 16 to 128 (standard: 64). The value $q = 7.2$ was identified through preliminary grid search on a held-out validation period.

At $q = 0.5$, the walker readily explores distant neighborhoods, potentially capturing broader industry relationships. At $q = 10.0$, the walker is heavily constrained to local neighborhoods, potentially triangulating tight sub-industry clusters. Intermediate values like $q = 7.2$ balance these strategies. Our experiments reveal that this balanced approach achieves optimal performance, however, further investigation into the most optimal parameters should be carried out.

3.5 Evaluation Metrics

For every tested configuration, we employed the following metrics:

1. **Sub-Industry Hit Rate (Primary):** For each stock, we compute the 3 nearest neighbors in embedding space (cosine similarity) and measure what fraction share the stock's GICS Sub-Industry classification. This is averaged across all stocks in the universe. Hit rates are reported as percentages. For context, with approximately 3 stocks per sub-industry on average, a random baseline would achieve approximately 2% hit rate.

2. **Gini Score:** Measures inequality in PageRank centrality distribution. High Gini indicates hub-dominated network structures where hubs are nodes with significantly higher degrees than nodes connected to them.

3. **Modularity:** Quantifies strength of community structure using Louvain community detection [11]. High modularity (> 0.7) indicates well-separated clusters.

4. **Assortativity:** Measures tendency of similar-degree nodes to connect. Positive assortativity suggests healthy mesh topology; negative suggests hub-and-spoke fragility.

5. **Cliff Analysis:** Identifies nodes whose nearest neighbor has cosine similarity below a critical threshold, indicating semantic isolation. Reported as percentage of nodes affected.

6. **Hubness:** Examines whether any single node connects to more than 25% of the graph, which can indicate sector leakage where unrelated stocks collapse toward a dominant hub (e.g., railways, banks).

4 Results

4.1 Overall Performance Summary

Across 114 experimental configurations spanning three universes (S&P 500, Russell 1000, Russell 3000), four training periods (6 months to 5 years), multiple Node2Vec parameters, and various graph filtering approaches, we find:

- **Maximum Hit Rate:** 56.13% (Russell 3000, 5-year period, $q = 7.2$, threshold $t = 0.65$)
- **S&P 500 Best:** 47.53% (5-year period, $q = 10.0$, $t = 0.75$)
- **Russell 1000 Best:** 54.56% (5-year period, $q = 10.0$, $t = 0.72$)
- **Russell 3000 Best:** 56.13% (5-year period, $q = 7.2$, $t = 0.65$)

To contextualize these results: the random baseline for this 163-class task (with approximately 3 stocks per sub-industry) is roughly 2%, making our 56% hit rate represent substantial learned signal. Prior work has primarily targeted the easier 11-sector classification task; Dolphin et al. (2022) achieved approximately 60% accuracy at this coarser granularity [3]. Direct comparison is difficult given the 14× difference in class granularity, but our results demonstrate that meaningful sub-industry structure can be recovered from correlation-based embeddings.

4.2 Critical Finding: Training Period Dominates

A significant discovery in our experiments is the importance of training period length. Table 1 shows hit rate performance by training window:

Table 1: Training Period Effect on Sub-Industry Hit Rate

Training Period	Mean Hit Rate	Max Hit Rate	N Configs
6 months	38.07%	39.87%	9
1 year	40.68%	43.54%	11
2 years	44.37%	49.62%	26
5 years	**51.81%**	**56.13%**	33

The 5-year training window shows a +7.4 percentage point improvement in mean hit rate compared to 2-year windows, and a +13.7 pp improvement over 6-month windows. This effect exceeds the impact of parameter optimization or universe expansion, suggesting that sub-industry correlation patterns require substantial historical depth to stabilize. This finding aligns with portfolio optimization literature indicating that stable covariance estimation requires multi-year samples [12, 19].

Short-term correlation matrices (6 months to 1 year) may be dominated by transient market conditions, macroeconomic shocks, or momentum effects that obscure fundamental industry relationships [9, 20]. Thus, they can be used to detect more recent, less noticeable trends, changes in nature of stocks, or sector rotations. Extended training periods (5 years) allow these temporary correlations to average out, revealing the underlying structural similarities between stocks in the same sub-industry.

4.3 Universe Expansion Effect

When controlling for training period (2 years) and using standard parameters ($q = 0.5$, $t = 0.65$), we observe consistent improvement with universe expansion:

Table 2: Universe Size Effect (2-Year Training Period, $q = 0.5$, $t = 0.65$)

Universe	**Constituents**	**Hit Rate**	**vs. S&P 500**
S&P 500	491	42.13%	baseline
Russell 1000	961	46.37%	+4.24 pp
Russell 3000	2682	46.37%	+4.24 pp

At 5-year training periods, the universe expansion effect becomes more pronounced:

Table 3: Universe Size Effect (5-Year Training Period, $q = 0.5$, $t = 0.65$)

Universe	**Constituents**	**Hit Rate**	**vs. S&P 500**
S&P 500	471	43.58%	baseline
Russell 1000	879	52.20%	+8.62 pp
Russell 3000	2177	55.47%	+11.89 pp

This pattern is consistent with the "Mid-Cap Bridge" hypothesis: mid-cap stocks may serve as topological intermediaries that increase the effective distance between weakly related large-cap stocks, thereby enhancing embedding resolution. The effect is nearly tripled at 5-year training periods (11.89 pp vs 4.24 pp at 2 years), suggesting that mid-cap bridging benefits compound with temporal stability.

However, we note an important caveat: the universe expansion effect cannot be cleanly separated from improved covariance estimation with more data points. Larger universes provide more off-diagonal correlation pairs, potentially yielding more stable correlation estimates [19]. Future work could disentangle these effects through matched sampling experiments.

4.4 Node2Vec Parameter Analysis

Table 4 summarizes performance across different q values:

Table 4: Node2Vec q Parameter Effect on Hit Rate

q **Value**	**Mean Hit Rate**	**Max Hit Rate**	**N Configs**
$q = 0.5$	43.23%	55.89%	45
$q = 7.2$	48.22%	**56.13%**	5
$q = 10.0$	49.19%	56.03%	14
$q = 20.0$	53.04%	53.04%	1

Key Findings:

- $q = 7.2$ (Intermediate/Balanced) achieves the highest absolute hit rate (56.13%), suggesting that balanced exploration-saturation strategies are optimal for sub-industry resolution.

- $q = 10.0$ (BFS-like) shows strong mean performance (49.19%) and the second-highest maximum (56.03%), indicating robust performance across configurations.
- $q = 0.5$ (DFS-like) shows high variance (mean 43.23%, max 55.89%), suggesting sensitivity to other hyperparameters.

4.5 Systematic Ablation Studies

To validate our core findings and isolate the effects of individual hyperparameters, we conducted additional ablation experiments systematically varying q values, thresholds, dimensions, and universes while holding other factors constant.

Clean Parameter Comparisons Across Universes. We performed matched comparisons at 5-year training periods to isolate universe effects from parameter choices. At $q = 0.5$ with $t = 0.65$, we observe: S&P 500 (43.6%) → R1000 (52.2%) → R3000 (55.5%), confirming consistent universe expansion benefits. Similarly, at $q = 10.0$ with $t = 0.65$: S&P 500 (44.3%) → R1000 (54.0%) → R3000 (56.0%). These controlled comparisons demonstrate that the mid-cap bridge effect persists across different walk strategies, with R3000 consistently outperforming smaller universes by 8–12 percentage points.

Threshold Sensitivity Analysis. We systematically varied correlation thresholds from $t = 0.55$ to $t = 0.75$ across universes at 5-year training periods. For R3000 with $q = 0.5$: $t = 0.55$ yields 53.9%, $t = 0.60$ yields 55.2%, $t = 0.65$ yields 55.5%, and $t = 0.70$ yields 55.9%. This shallow sensitivity curve indicates robustness to threshold selection within the 0.6–0.7 range, though performance degrades notably outside this window. The S&P 500 shows similar patterns but with overall lower hit rates, again confirming that universe size effects dominate threshold tuning.

Embedding Dimension Effects. We tested dimensions from 64 to 128 on 5-year R1000 and R3000 configurations. For R1000 at $q = 10.0$, $t = 0.72$: 64d achieves 54.6%, 96d achieves 54.6%, and 128d achieves 54.1%. For R3000 at $q = 0.5$, $t = 0.65$: 64d achieves 55.5% and 96d achieves 55.8%. These results suggest diminishing returns beyond 64 dimensions for this task, with marginal gains potentially offset by increased computational cost and overfitting risk.

Return Parameter Analysis. The return parameter p controls backtracking probability in random walks. We tested $p = 5.0$ (anti-return bias) versus the standard $p = 1.0$ on 5-year configurations. For R3000 with $q = 0.5$, $t = 0.65$: $p = 1.0$ yields 55.5% while $p = 5.0$ yields 55.7%, a negligible 0.2 pp difference. For R1000: $p = 1.0$ yields 52.2% while $p = 5.0$ yields 49.5%. These results indicate that p has minimal impact compared to q, training period, and universe size, suggesting that local versus global exploration (q) matters far more than backtracking behavior (p).

The Intermediate q Advantage. We tested $q = 7.2$ to probe the transition region between exploration ($q = 0.5$) and saturation ($q = 10.0$) strategies. At 5 years on R3000 with $t = 0.65$: $q = 7.2$ achieves **56.1%**, outperforming both $q = 0.5$ (55.5%) and $q = 10.0$ (56.0%). This represents our best overall performance and suggests that balanced random walk strategies—neither purely depth-first nor breadth-first—are optimal for capturing sub-industry relationships. The balanced approach may allow walkers to escape local minima while still maintaining sufficient neighborhood context.

4.6 Semantic Neighbor Analysis

Table 5 shows representative nearest neighbors from a Russell 1000 configuration (2-year training, $q = 7.2$, $t = 0.65$, hit rate $\approx 45\%$). These examples illustrate the types of relationships captured by the embedding space:

Table 5: Representative Nearest Neighbors (Top 5) from R1000 Configuration

Ticker	Neighbors (Top 5)	Semantic Interpretation
AAPL	QRVO, SWKS, CRUS, OLED, ALGM	**Supply Chain.** RF & Audio chip suppliers
JPM	C, WFC, FHN, WBS, BAC	**Banking Tier.** Global SIBs with Regional Banks
JNJ	BMY, ABT, GILD, MDT, ED	**Pharma Majors.** (Note: ED is utility leak)
XOM	CVX, EOG, APA, COP, OXY	**Energy Mesh.** Integrated Oil & E&P peers
TSLA	MPWR, LSCC, ON, QCOM, ENTG	**EV Power Systems.** Power management chips
AMD	SMCI, KLAC, AMAT, LRCX, ONTO	**Semi Mfg & Servers.** Capital equipment cluster
LOW	PHM, LEN.B, TOL, NVR, HD	**Home Construction.** Clusters with homebuilders
KO	PEP, MNST, MCD, KDP, CL	**Beverages.** Strong sub-industry cohesion
PEP	KDP, MNST, KO, PM, MCD	**Snack & Bev.** Reciprocal with KO
AMZN	GOOGL, GOOG, MSFT, ROKU, META	**Mega-Cap Tech.** Cloud/Ads peers

Note: The absolute best-performing configuration (R3000, 5y, $q = 7.2$, 56.13% hit rate) produces different neighbor patterns with more emphasis on large-cap mega-clusters (e.g., AAPL connects to TSLA, MSFT, GOOGL rather than supply chain partners). This suggests that optimal hit rates and semantic interpretability may involve different tradeoffs. Essentially, the choice of parameters depends on the specific use case.

4.7 Graph Topology Metrics

Our best-performing configurations exhibit strong structural properties:

- **Modularity:** The best configurations achieve modularity > 0.50, indicating well-separated community structure suitable for Node2Vec traversal
- **Assortativity:** Positive assortativity (0.5-0.7) in successful configs indicates mesh-like connectivity
- **Low Hubness:** Top hubs connect to $<10\%$ of nodes, avoiding sector leakage
- **Minimal Cliffs:** Only 13-19% of nodes are semantically isolated in best configs

Importantly, Forced Connectivity adds only 1-3% additional edges while reducing cliff percentage from 25-30% to 13-19%, demonstrating its effectiveness at graph repair without compromising signal quality.

4.8 Filtering Method Comparison

Simple Pearson correlation outperformed sophisticated alternatives:

- **Pearson + Threshold + Forced Connectivity:** Mean hit rate = 44.85%
- **Graphical Lasso:** Mean hit rate = 39.35%
- **MST:** Mean hit rate = 42.18%

This validates our core hypothesis that aggressive denoising through precision matrices or tree structures may discard meaningful sub-industry signals, consistent with observations that sparse precision estimators can be unstable in financial applications [16]. The simplicity and computational efficiency of correlation thresholding combined with targeted connectivity repair provides the best performance-to-complexity ratio.

5 Discussion

5.1 The Role of Historical Depth

The 7.4 pp improvement from 2-year to 5-year windows suggests that sub-industry correlation patterns are stable at multi-year timescales but noisy at shorter horizons. This has important implications:

- Short-term embeddings (6mo-1y) may reflect momentum, sentiment, or macroeconomic regime more than fundamental industry structure [20]

- 5-year windows average across multiple market cycles, revealing persistent structural relationships
- This finding aligns with portfolio construction practices that typically require 3-5 years of data for stable covariance estimation [12, 19]

5.2 The Mid-Cap Bridge Hypothesis

We hypothesized that mid-cap stocks improve resolution by serving as topological intermediaries. Our evidence shows consistent and substantial improvement with universe expansion. The mechanism appears strongest when combined with extended training periods, as the 2-year universe effect (4.2 pp) is less than half the 5-year effect (11.9 pp). This suggests that:

In an S&P 500-only graph, two loosely related large-cap stocks (e.g., AAPL and a home construction stock) may appear as neighbors simply because no intermediate nodes exist. When mid-cap regional technology distributors or specialized components suppliers are added, they become the preferred neighbors for AAPL, effectively "pushing apart" the unrelated large-cap vectors in the embedding space. This increases the angular distance between truly dissimilar stocks.

Over extended training periods, these topological relationships stabilize, allowing the mid-cap bridges to consistently separate dissimilar large-caps while connecting related ones through specialized intermediaries.

However, we emphasize an important caveat: the mid-cap bridge effect cannot be cleanly separated from improved covariance estimation with more data points. Larger universes provide more off-diagonal correlation pairs, potentially yielding more stable correlation estimates [19]. Future work could disentangle these effects through matched sampling experiments—e.g., comparing R3000 results to bootstrap samples of S&P 500 stocks expanded with random noise nodes.

5.3 Limitations and Future Work

Several limitations warrant acknowledgment:

1. **Temporal Stability:** We use static 5-year training windows ending at a fixed cutoff. Financial network topology varies substantially across market regimes [8, 20], and our findings may not generalize to periods of market stress or structural change. Rolling window validation across multiple market cycles would strengthen these conclusions.
2. **Causality:** While our controlled ablations are consistent with the mid-cap bridge hypothesis, definitive causal proof would require synthetic experiments or quasi-experimental designs exploiting universe additions/deletions.
3. **Supply Chain Validation:** The AAPL $\rightarrow$ QRVO/SWKS neighbors suggest supply chain detection, but we have not validated these relationships against actual supplier databases (e.g., FactSet supply chain data).
4. **Optimal q Search:** We tested discrete values (0.5, 7.2, 10.0, 20.0). Finer grid search around $q \in [5.0, 10.0]$ might reveal further improvements.
5. **Statistical Significance:** While we report systematic improvements across configurations, formal statistical tests (e.g., bootstrap confidence intervals, paired comparisons) would strengthen claims about the reliability of observed differences.

6 Conclusion

This study demonstrates that sub-industry resolution in financial embeddings is achievable through three key elements: (1) extended training periods capturing stable multi-year correlation patterns, (2) universe expansion providing topological intermediaries, and (3) a novel Forced Connectivity graph repair mechanism that preserves signal quality while ensuring navigability.

Our maximum hit rate of 56.13% at the 163-class GICS Sub-Industry granularity—substantially above the approximately 2% random baseline—represents meaningful progress on an under-explored problem. Although prior literature targets broad 11-sector classification, we demonstrate that Node2Vec-based embeddings can capture finer economic relationships, though substantial room for improvement remains.

The Forced Connectivity mechanism represents a generalizable contribution beyond financial networks: any correlation network construction task facing the threshold-connectivity tradeoff could benefit from this two-stage approach.

Importantly, we highlight several findings that challenge conventional assumptions:

1. Training period selection—often treated as a mundane preprocessing choice—has the largest impact on embedding quality, exceeding the effects of sophisticated parameter optimization
2. Mid-cap inclusion correlates with substantial benefits (11.9 pp gain), suggesting that universe design matters as much as algorithm design, though this effect may be partially confounded with improved covariance estimation
3. Intermediate exploration parameters ($q = 7.2$) outperform extreme values, indicating that balanced strategies are optimal

These findings suggest that future research should systematically explore data characteristics (temporal window, universe composition) before investing heavily in algorithmic complexity.

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